

1. Terms and experimental design

Research question: What are the factors, and which comparisons do they define?

Method: Define each factor and list its levels before interpreting the models.

Finding: Each generator has seven conditions, evaluated with two training modes and two seeds. Real-only runs are separate reference points.

Term	Meaning	Levels used
dataset	Task being evaluated.	Classification: Adult, Census KDD, Covertypes, MNIST12, MNIST28. Regression: News and Housing.
generator	Model producing synthetic rows.	CTGAN; TVAE.
source / has_y	Columns used to fit the generator.	full / has_y=1: (X,Y). xonly / has_y=0: X. Both generate features X.
labeler	Predictor trained on real training data to assign synthetic Y.	RF (random forest); XGB (XGBoost); DNN (neural network).
condition	One complete choice of feature source and target construction.	generated; rf_full; rf_xonly; xgb_full; xgb_xonly; dnn_full; dnn_xonly.
approach	How the synthetic target is obtained.	Generated Y; relabeled Y (the six RF/XGB/DNN conditions).
group / approach_has_y	Three groups used in the mixed model.	generated; hybrid_xonly; hybrid_full. Each hybrid group averages RF/XGB/DNN equally.
mode	Rows used for downstream training.	synthetic: 100,000 synthetic rows only. mix: all real training rows plus 100,000 synthetic rows.
seed	Experiment random-number setting.	42; 43. The prepared split is the same for both.
original	Real-only reference model.	Real training rows only; outside the seven-condition ANOVA.

X = predictors; **Y** = target. Generated Y exists only for a full-table generator; there is no generated-target X-only condition. The downstream evaluator is a neural model in every arm. RF/XGB are labelers, not downstream evaluators.

These are **factor levels**, not target class labels. The selected snapshot has 406 runs: 290 classification and 116 regression. Classification models use macro F1; regression metrics are kept separate.

Macro F1 averages the F1 scores of all target classes equally, on a 0-1 scale.

2. Does relabeling improve downstream utility?

Research question: Do targets predicted from real training data outperform the generator's own targets?

Method: Compare the mean of all six relabeled conditions with generated Y. Use a dataset-intercept mixed model, then check the contrast using five dataset means.

Finding: Mean macro F1 rises from **0.7090 to 0.8010: +0.0920 points (+12.98% relative)**. The model finds a clear effect; evidence across datasets is weaker after adjustment.

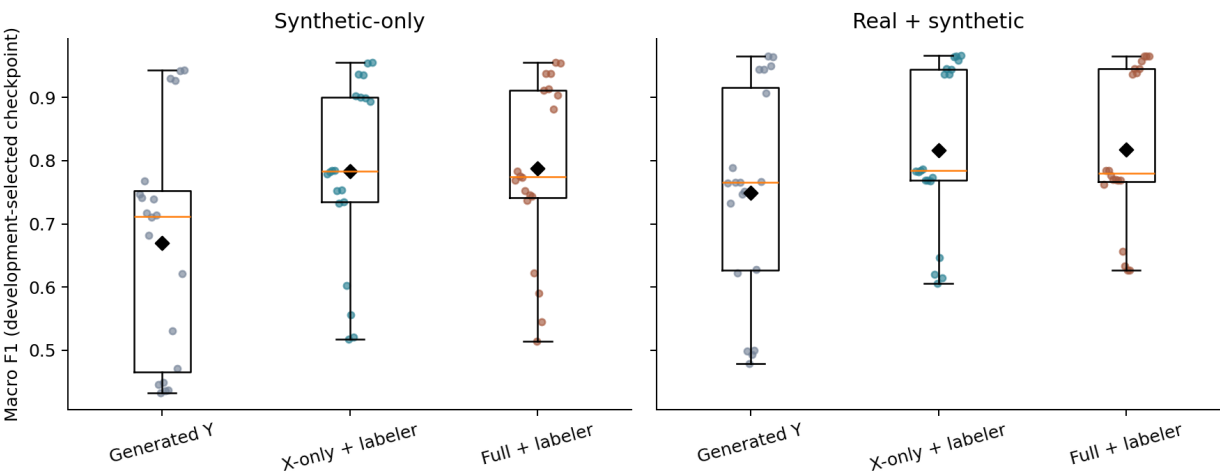
Average effect by training mode

Training mode	Generated	Relabeled	F1 change	Relative gain
pooled	0.7090	0.8010	+0.0920	12.98%
synthetic	0.6692	0.7855	+0.1163	17.38%
mix	0.7488	0.8166	+0.0677	9.04%

Same pooled contrast, two scopes of inference

Test and question answered	95% CI for F1 change	Adjusted p	Sig.
Mixed model: average effect under its dataset-intercept assumptions.	[+0.0664, +0.1176]	7.47e-12	***
Dataset paired t-test: how variable is the effect across five tasks?	[+0.0185, +0.1656]	0.0924	.

Significance: $p < 0.001 = ***$; $p < 0.01 = **$; $p < 0.05 = *$; $p < 0.10 = .$; blank otherwise. Bold rows have $p < 0.05$. Symbols use Holm-adjusted p over four planned contrasts.



Boxes show variation in macro F1 across tasks, seeds and generators. Black diamonds are group means.

The dataset-level raw t-test p is 0.0255; exact sign-flip p is 0.0625. The confidence intervals are unadjusted; Holm p accounts for testing four contrasts. Five positive task effects do not establish improvement on every future dataset. Relative gain = score change / generated mean; +0.0920 F1 equals +9.20 percentage points.

3. Does including Y in generator fitting help?

Research question: Among relabeled constructions, is fitting on (X,Y) better than fitting on X only?

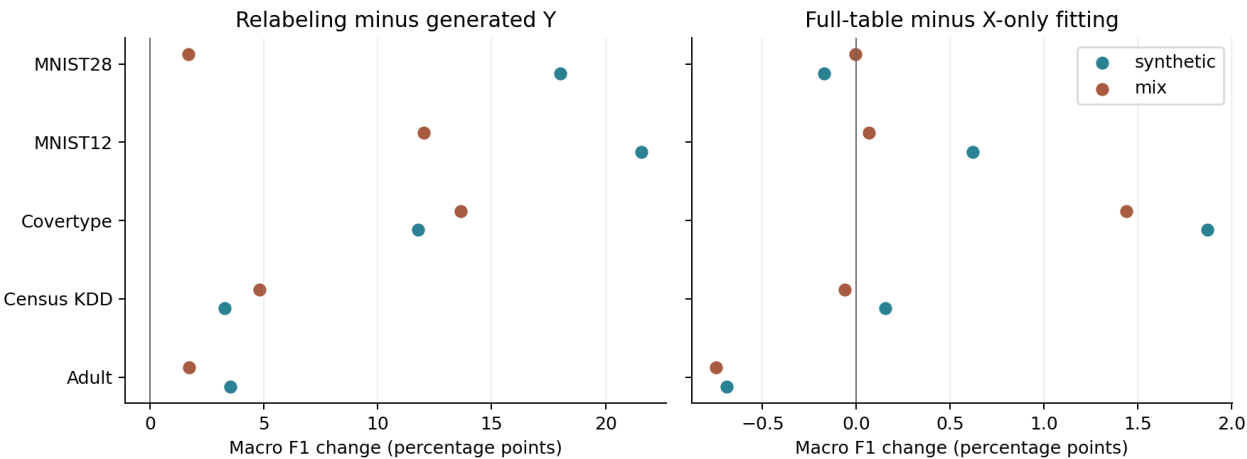
Method: Match full and X-only results within dataset, seed, generator, labeler and mode; report their mean difference and dataset-level confidence interval.

Finding: The pooled change is **+0.00248 macro-F1 points (+0.31% relative)**. The confidence interval includes zero; there is no clear full-table advantage.

Training	X-only	Full	F1 change	95% paired-t CI	Adjusted p	Sig.
pooled	0.7998	0.8023	+0.0025	[-0.0084, +0.0134]	0.5621	
synthetic	0.7837	0.7873	+0.0036	[-0.0085, +0.0156]	0.4575	
mix	0.8159	0.8172	+0.0014	[-0.0085, +0.0113]	0.7182	

Significance: $p < 0.001 = ***$; $p < 0.01 = **$; $p < 0.05 = *$; $p < 0.10 = .$; blank otherwise. Bold rows have $p < 0.05$. Symbols use Holm-adjusted dataset-level p over four contrasts.

Dataset	Relabeled - generated	Full - X-only
Adult	+0.0262	-0.0072
Census KDD	+0.0404	+0.0005
Covertypes	+0.1269	+0.0165
MNIST12	+0.1680	+0.0034
MNIST28	+0.0986	-0.0009



Full and X-only both generate X and replace Y with the same type of labeler. This question concerns including Y during generator fitting. A nonsignificant difference does not prove the two constructions equivalent.

4. Do method effects differ across datasets?

Research question: Can one common method effect describe all five classification tasks?

Method: Fit the original-project blocked model, then add dataset x condition. These ordinary linear models are exploratory because scores within a dataset are correlated.

Finding: The dataset x condition interaction is strong in the exploratory model ($p = 3.76e-12$). Method benefits vary by task, so one pooled effect is incomplete.

Blocked model: macro F1 ~ dataset + generator * condition * mode.

Term	SS	df	F	Nominal p	Sig.
dataset	4.5354	4	341.62	1.38e-99	***
generator	0.0204	1	6.14	0.0138	*
condition	0.3840	6	19.28	1.97e-18	***
mode	0.1009	1	30.41	8.78e-08	***
generator x condition	0.1063	6	5.34	3.35e-05	***
generator x mode	0.0043	1	1.29	0.2563	
condition x mode	0.0212	6	1.07	0.3841	
generator x condition x mode	0.0188	6	0.94	0.4642	
Residual	0.8231	248			

Dataset-interaction model: macro F1 ~ dataset * condition + generator + mode.

Term	SS	df	F	Nominal p	Sig.
dataset	4.5354	4	428.58	8.82e-109	***
condition	0.3840	6	24.19	2.10e-22	***
generator	0.0204	1	7.71	0.0059	**
mode	0.1009	1	38.15	2.74e-09	***
dataset x condition	0.3308	24	5.21	3.76e-12	***
Residual	0.6429	243			

Significance: $p < 0.001 = ***$; $p < 0.01 = **$; $p < 0.05 = *$; $p < 0.10 = .$; blank otherwise. Bold rows have $p < 0.05$. Symbols use nominal p; these exploratory tables are not adjusted.

How to read the terms: generator asks whether CTGAN and TVAE differ on average. condition asks whether the seven constructions differ. mode asks whether synthetic-only and mix differ. dataset x condition asks whether the construction effect changes with the dataset. Other x terms ask the same dependence question for their named factors.

In the formulas, **C(term)** means categorical. **a * b** includes a, b and their interaction; **a:b** is the interaction alone. **SS** is variation assigned to a term; **df** counts its independent comparisons; **F** compares term variation with residual variation. **Residual** is the variation the model leaves unexplained.

5. Which generator and labeler combinations benefit?

Research question: Which relabeled construction has the largest observed improvement over its own generator baseline?

Method: Compare each generator/labeler/source with its matched generated-Y baseline, averaging seeds and giving each of the five tasks equal weight.

Finding: CTGAN has larger observed gains than TVAE; DNN gives the largest observed gain for both generators. Individual combinations remain exploratory: none passes Holm adjustment across 24 tests.

Synthetic-only

Generator	Labeler	Source	F1 change	Relative gain	Adjusted p	Sig.
CTGAN	RF	full	+0.1774	30.1%	1.0000	
CTGAN	RF	xonly	+0.1778	30.2%	1.0000	
CTGAN	XGB	full	+0.1860	31.6%	1.0000	
CTGAN	XGB	xonly	+0.1772	30.1%	1.0000	
CTGAN	DNN	full	+0.2287	38.8%	1.0000	
CTGAN	DNN	xonly	+0.2239	38.0%	1.0000	
TVAE	RF	full	+0.0219	2.9%	1.0000	
TVAE	RF	xonly	+0.0184	2.5%	1.0000	
TVAE	XGB	full	+0.0275	3.7%	0.3251	
TVAE	XGB	xonly	+0.0254	3.4%	0.6707	
TVAE	DNN	full	+0.0672	9.0%	1.0000	
TVAE	DNN	xonly	+0.0645	8.6%	1.0000	

Real + synthetic

Generator	Labeler	Source	F1 change	Relative gain	Adjusted p	Sig.
CTGAN	RF	full	+0.0793	11.1%	1.0000	
CTGAN	RF	xonly	+0.0857	12.0%	1.0000	
CTGAN	XGB	full	+0.0928	13.0%	1.0000	
CTGAN	XGB	xonly	+0.0902	12.6%	1.0000	
CTGAN	DNN	full	+0.1277	17.8%	1.0000	
CTGAN	DNN	xonly	+0.1243	17.3%	1.0000	
TVAE	RF	full	+0.0189	2.4%	1.0000	
TVAE	RF	xonly	+0.0160	2.1%	1.0000	
TVAE	XGB	full	+0.0325	4.2%	1.0000	
TVAE	XGB	xonly	+0.0255	3.3%	1.0000	
TVAE	DNN	full	+0.0594	7.6%	1.0000	
TVAE	DNN	xonly	+0.0605	7.7%	1.0000	

Significance: $p < 0.001 = ***$; $p < 0.01 = **$; $p < 0.05 = *$; $p < 0.10 = .$; blank otherwise. Bold rows have $p < 0.05$. Symbols use Holm-adjusted dataset-level p over 24 combinations.

These are gains over generated Y, not a direct significance test of DNN versus RF/XGB. No best seed or test epoch is selected. Per-combination means, confidence intervals and raw p-values remain in `labeler_effects.csv`.

6. Do the conclusions extend to regression?

Research question: How do relabeled tables compare with generated-target tables for News and Housing?

Method: Report separate means for R2, D2 absolute error and saved NMAE. Average seeds and generators equally, and average RF/XGB/DNN within each hybrid group.

Finding: Housing relabeling improves all three metrics over generated Y. News improves R2 slightly, while its absolute-error metrics worsen slightly. The two regression tasks are descriptive comparisons.

Metric	Meaning	Better direction		
R2	Squared-error reduction against a constant test-mean baseline.	Higher		
D2 absolute error	1 - MAE / MAE of the constant test-median baseline.	Higher		
NMAE	MAE / standard deviation of real test targets (ddof=0).	Lower		

Dataset / training	Construction	R2	D2	NMAE
News / synthetic	Real only	-0.0290	-0.0445	0.2686
News / synthetic	Generated Y	-0.0235	0.0099	0.2546
News / synthetic	Full hybrid	-0.0004	-0.0060	0.2587
News / synthetic	X-only hybrid	0.0027	-0.0098	0.2597
News / mix	Real only	-0.0290	-0.0445	0.2686
News / mix	Generated Y	-0.0196	0.0300	0.2494
News / mix	Full hybrid	-0.0008	0.0104	0.2544
News / mix	X-only hybrid	0.0002	0.0079	0.2551
Housing / synthetic	Real only	0.5125	0.2822	0.5507
Housing / synthetic	Generated Y	0.6730	0.4595	0.4146
Housing / synthetic	Full hybrid	0.7930	0.6061	0.3022
Housing / synthetic	X-only hybrid	0.7946	0.6025	0.3049
Housing / mix	Real only	0.5125	0.2822	0.5507
Housing / mix	Generated Y	0.6676	0.4607	0.4137
Housing / mix	Full hybrid	0.7872	0.6022	0.3052
Housing / mix	X-only hybrid	0.7905	0.5956	0.3103

R2 and D2 can be negative, so relative percentage changes in those scores are omitted. Saved test normalization and D2 sidecars are used directly. There are no regression ANOVA significance stars: two tasks do not support the same across-dataset inference as the five-task classification analysis.

Appendix A. Why include a repeated-measures ANOVA?

Research question: Are factor effects detectable relative to seed-to-seed variation on these fixed datasets?

Method: Treat dataset as fixed and dataset/seed as the repeat unit. Compare all generator/condition/mode scores within each unit; partition condition into approach, labeler and has_y.

Finding: Relabeling is significant within this fixed-task model (adjusted p = **0.0043**); full versus X-only is not. This is a supporting check, not another test of generalization to new tasks.

Effect	F	Adjusted df	GG p	Holm p	Sig.
generator	18.02	1.00, 5.00	0.0081	0.2103	
approach	114.52	1.00, 5.00	0.0001	0.0043	**
labeler	691.21	1.19, 5.94	1.75e-07	6.48e-06	***
has_y	4.12	1.00, 5.00	0.0981	1.0000	
mode	452.26	1.00, 5.00	4.26e-06	0.0002	***
dataset x approach	9.49	4.00, 5.00	0.0148	0.3408	
dataset x labeler	426.99	4.76, 5.94	1.69e-07	6.44e-06	***
dataset x has_y	10.34	4.00, 5.00	0.0123	0.2962	
approach x mode	53.33	1.00, 5.00	0.0008	0.0249	*
generator x approach	37.60	1.00, 5.00	0.0017	0.0502	.

Significance: p < 0.001 = ***; p < 0.01 = **; p < 0.05 = *; p < 0.10 = .; blank otherwise. Bold rows have p < 0.05. Symbols use Holm p over all 39 repeated-measures terms.

GG (Greenhouse-Geisser) adjusts for unequal covariance among repeated comparisons. Adjusted df lists numerator and denominator degrees of freedom. **approach** compares generated Y with the six hybrids; **labeler** and **has_y** apply within hybrids only.

Analysis	What it contributes	Location
Blocked / interaction models	Show which patterns vary with factors and datasets. Their ordinary p-values are exploratory.	Section 4
Mixed model	Estimates average construction effects with a dataset random intercept; its covariance assumptions matter.	Section 2
Dataset-level contrasts	Use five task means to assess how effects vary across tasks.	Sections 2-3
Repeated-measures ANOVA	Tests fixed-task effects against seed variation. It does not create more independent datasets.	This appendix

There are ten dataset/seed units and five residual seed degrees of freedom. Both seeds share the holdout, so these tests do not measure test-sample uncertainty. The full 39-term table and partial eta-squared values are retained in repeated_anova.csv.

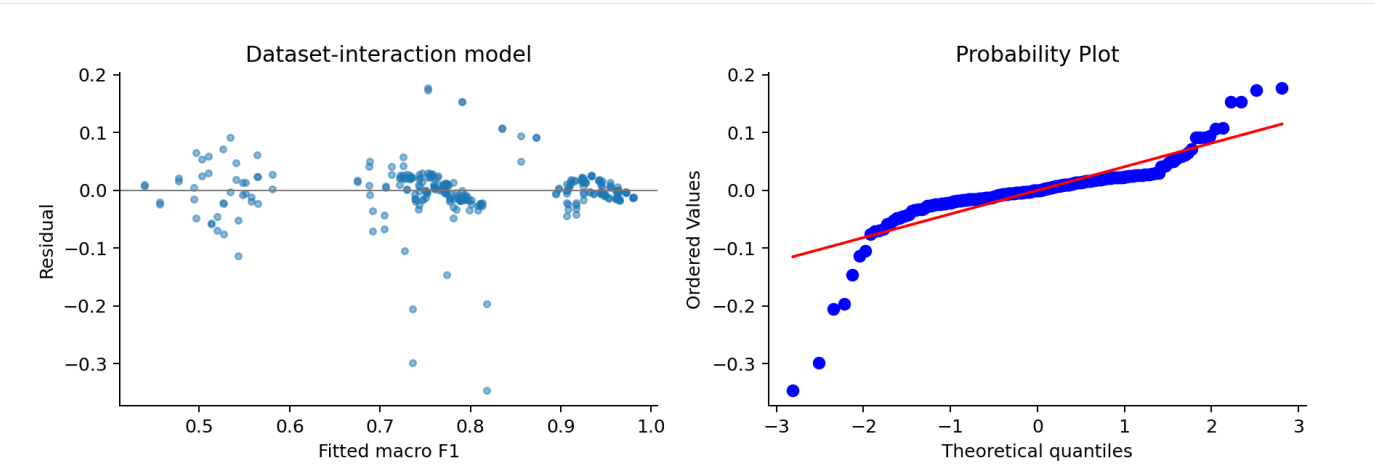
Appendix B. Why keep the model diagnostics?

Research question: Do the original-project models adequately describe the score patterns?

Method: Compare full-interaction, blocked and dataset-interaction fits; inspect residual-versus-fitted and normal Q-Q plots.

Finding: The full model fits closely, but simpler-model residuals are structured and nonnormal. Their nominal ANOVA p-values are insufficient on their own.

Model	What it allows	R2	Adjusted R2	Residual df
Full interaction	All dataset/generator/condition/mode interactions.	0.9890	0.9780	140
Blocked	Dataset starting scores; generator/condition/mode interactions.	0.8631	0.8460	248
Dataset interaction	Dataset-specific condition effects; generator and mode main effects.	0.8931	0.8773	243



Dataset-interaction model residuals: pattern in the left plot and departure from normality in the right plot.

Model 3 Shapiro $p = 3.50e-21$. Macro F1 stays on its original 0-1 scale; no transformation was chosen to improve significance. Model summaries and the full-interaction ANOVA remain in the package.

Mixed-model specification: $\text{macro F1} \sim \text{group} * \text{generator} * \text{mode} + (1 | \text{dataset})$, fitted to 120 group means. REML estimates the variance components; Wald intervals use a normal approximation. The fit converged. A dataset random intercept allows different starting scores, but no dataset-specific treatment slopes. The dataset-level contrast checks that limitation.

Appendix C. Sources, exclusions and reproducibility

Research question: Are the report values traceable to completed runs with the intended settings?

Method: Use the verified frozen records, completed-record aggregate and source-linked D2 sidecars. Preserve all statistical values while revising the report presentation.

Finding: All 406 selected runs and 116 D2 sidecars verified. This revision changes explanations, ordering and table styling; it does not rerun experiments or statistical models.

Dataset	Source namespace	Runs
Adult	corrected_v2	58 / 58
Covertypes	corrected_v2	58 / 58
Census KDD	census_kdd_weighted_macro_f1_20261005	58 / 58
MNIST12	mnist_head_fixed_20261009	58 / 58
MNIST28	mnist_head_fixed_20261009	58 / 58
News	news_log_v1	58 / 58
Housing	housing_no_faker_20261009	58 / 58

Source snapshot: October 10, 2026, 1:08:42 a.m. Chicago. D2 was added from the verified 1:10 a.m. derived-metric update. These are the same frozen results used in the earlier report.

Scope: CTGAN/TVAE and RF/XGB/DNN; seeds 42/43; synthetic-only/mix. Credit, Intrusion, NB/PCA-GMM, exploratory dropout trials and simulated data are excluded. Completion here does not establish completion of the broader matrix.

Checks and caveats: models use saved development-selected checkpoints, not test-epoch maxima. Generator budgets remain 500 epochs / batch 500 / 100,000 rows. Splits are shared across seeds. MNIST uses the repaired output heads; its retained MNIST12 TVAE tables still carry the documented holdout-feature collision caveat.

File	Purpose
real_data_328_analysis.Rmd	Editable report and optional equivalent R model specification.
planned_contrasts.csv / mixed_contrasts.csv	Dataset-level and mixed-model effects, confidence intervals and p-values.
labeler_effects.csv / repeated_anova.csv	All method contrasts and repeated-measures terms.
scores.csv / source_snapshot.json	Saved outcomes and source-record hashes.
regression_* / d2_snapshot.zip	Regression scores/effects and verified D2 sidecars.

Numerical fits ran remotely in Python. The optional R refit code has not been executed; lmerTest degrees of freedom can differ from the reported normal-Wald intervals.